Chapter 13. Classification of Covering Spaces

Equivalence of coverings, universal covering space, covering transformations, existence of coverings

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Throughout this chapter a covering map $p : E \to B$ is assumed to have E and B path connected and locally path connected. For $p(e_0) = b_0$ let $H_0 = p_*(\pi_1(E, e_0))$, a subgroup of $\pi_1(B, b_0)$ isomorphic to $\pi_1(E, e_0)$ (Theorem 9.17).

13.1 Equivalence of covering spaces

Covering maps $p : E \to B$ and $p' : E' \to B$ are **equivalent** if there is a homeomorphism $h : E \to E'$ with $p = p' \circ h$ (an **equivalence**).

Lemma 13.1 (79.1, general lifting lemma)

Let $p : E \to B$ be a covering map, $p(e_0) = b_0$, and $f : Y \to B$ continuous with $f(y_0) = b_0$, where Y is path connected and locally path connected. Then f has a lifting $\tilde{f} : Y \to E$ with $\tilde{f}(y_0) = e_0$ iff $f_*(\pi_1(Y, y_0)) \subset p_*(\pi_1(E, e_0))$. Such a lifting is unique.

Proof. If $\tilde{f}$ exists, $f_* = p_*\tilde{f}_*$. Uniqueness: for $y_1 \in Y$ and a path α from y_0 to y_1, $\tilde{f} \circ \alpha$ is the lifting of $f \circ \alpha$ from e_0, so $\tilde{f}(y_1)$ is its end point. Existence: define $\tilde{f}(y_1)$ as the end point of the lifting of $f \circ \alpha$ from e_0. It is well defined: if β is another path from y_0 to y_1, lift $f \circ \alpha$ to γ from e_0 and $f \circ \bar{\beta}$ to δ from $\gamma(1)$; then $\gamma * \delta$ lifts the loop $f \circ (\alpha * \bar{\beta})$, whose class lies in $f_*\pi_1(Y, y_0) \subset H_0$, so it is a loop (Theorem 9.17(c)); then $\bar{\delta}$ is the lifting of $f \circ \beta$ from e_0 and ends at $\gamma(1)$. Continuity at y_1: given a neighbourhood N of $\tilde{f}(y_1)$, choose a path-connected evenly covered $U \ni f(y_1)$ whose slice V_0 through $\tilde{f}(y_1)$ lies in N, and a path-connected neighbourhood W of y_1 with $f(W) \subset U$. For $y \in W$ and a path δ in W from y_1 to y, the lifting of $f \circ (\alpha * \delta)$ is $\gamma * ((p|V_0)^{-1} \circ f \circ \delta)$, ending in V_0. So $\tilde{f}(W) \subset V_0 \subset N$. ∎

Theorem 13.2 (79.2)

Let $p : E \to B$, $p' : E' \to B$ be covering maps with $p(e_0) = p'(e_0') = b_0$. There is an equivalence $h : E \to E'$ with $h(e_0) = e_0'$ iff $H_0 = p_*\pi_1(E, e_0)$ and $H_0' = p_*'\pi_1(E', e_0')$ are equal. Such h is unique.

Proof. If h exists, h_* is an isomorphism and $p = p'h$ gives $H_0 = H_0'$. Conversely, Lemma 13.1 gives liftings $h : E \to E'$ of p with $h(e_0) = e_0'$ and $k : E' \to E$ of p' with $k(e_0') = e_0$. Then $k \circ h$ and id_E both lift p and fix e_0, so they are equal by uniqueness; likewise $h \circ k = \mathrm{id}$. Uniqueness of h is that of liftings. ∎

Lemma 13.3 (79.3)

Let $e_0, e_1 \in p^{-1}(b_0)$ and $H_i = p_*\pi_1(E, e_i)$. (a) If γ is a path in E from e_0 to e_1 and $\alpha = p \circ \gamma$, then $[\alpha]H_1[\alpha]^{-1} = H_0$. (b) Conversely, if H is conjugate to H_0, then $H = H_1$ for some $e_1 \in p^{-1}(b_0)$.

Proof. (a) If $[h] \in H_1$, $h = p \circ \tilde{h}$ with $\tilde{h}$ a loop at e_1; $\gamma * \tilde{h} * \bar{\gamma}$ is a loop at e_0 mapping to $[\alpha][h][\alpha]^{-1}$, so $[\alpha]H_1[\alpha]^{-1} \subset H_0$. Using $\bar{\gamma}$, $[\alpha]^{-1}H_0[\alpha] \subset H_1$. (b) If $H_0 = [\alpha]H[\alpha]^{-1}$, lift α to γ from e_0 and let $e_1 = \gamma(1)$; by (a), $H_0 = [\alpha]H_1[\alpha]^{-1}$, so $H = H_1$. ∎

Theorem 13.4 (79.4)

p and p' are equivalent iff H_0 and H_0' are conjugate in $\pi_1(B, b_0)$.

Proof. If h is an equivalence and $e_1' = h(e_0)$, then $H_0 = p_*'\pi_1(E', e_1')$ by Theorem 13.2, which is conjugate to H_0' by Lemma 13.3. Conversely, if they are conjugate, Lemma 13.3(b) gives e_1' with $p_*'\pi_1(E', e_1') = H_0$, and Theorem 13.2 gives an equivalence carrying e_0 to e_1'. ∎

Example 13.1

$\pi_1(S^1) \cong \mathbb{Z}$ is abelian with subgroups $n\mathbb{Z}$. So every covering of S^1 is equivalent to $p : \mathbb{R} \to S^1$ (trivial subgroup) or to $z \mapsto z^n$ (subgroup $n\mathbb{Z}$).

13.2 The universal covering space

If E is simply connected, p is a **universal covering**; by Theorem 13.4 it is unique up to equivalence.

Lemma 13.5 (80.1)

Let B be path connected and locally path connected, and $p : E \to B$ a covering map with E not necessarily path connected. If E_0 is a path component of E, then $p|E_0 : E_0 \to B$ is a covering map.

Proof. E is locally path connected, so E_0 is open and $p(E_0)$ is open. If $x \in \overline{p(E_0)}$, take a path-connected evenly covered $U \ni x$; some slice V meets E_0, and being path connected lies in E_0, so $x \in U = p(V) \subset p(E_0)$. Thus $p(E_0) = B$. For any such U, $(p|E_0)^{-1}(U)$ is the union of the slices meeting E_0, each contained in E_0 and mapped homeomorphically onto U. ∎

Lemma 13.6 (80.2)

Let $p = r \circ q$ with $p : X \to Z$, $q : X \to Y$, $r : Y \to Z$ continuous (all spaces path connected and locally path connected). (a) If p and r are covering maps, so is q. (b) If p and q are covering maps, so is r.

Proof. Let $x_0 \in X$, $y_0 = q(x_0)$. (a) q is surjective: for a path $\tilde{\alpha}$ in Y from y_0 to y, lift $r \circ \tilde{\alpha}$ through p to $\tilde{\tilde{\alpha}}$ from x_0; $q \circ \tilde{\tilde{\alpha}}$ lifts $r \circ \tilde{\alpha}$ through r from y_0, so equals $\tilde{\alpha}$, and y is in the image. Given y, take a path-connected $U \ni r(y)$ evenly covered by both p and r, and the slice V of $r^{-1}(U)$ containing y. Each slice U_α of $p^{-1}(U)$ is connected, so q maps it into one slice of $r^{-1}(U)$; $q^{-1}(V)$ is the union of those U_α mapped into V, and on each $q = (r|V)^{-1} \circ (p|U_\alpha)$ is a homeomorphism onto V. (b) Let U be a path-connected neighbourhood of z evenly covered by p, and V_β the path components of $r^{-1}(U)$, open and disjoint. The slices U_α of $p^{-1}(U)$ are the path components of $p^{-1}(U)$, and q maps each into one V_β; so $q^{-1}(V_\beta)$ is a union of slices and, q being surjective, contains some U_α. By Theorem 9.10, q restricts to a covering map $q^{-1}(V_\beta) \to V_\beta$, and U_α is a path component of $q^{-1}(V_\beta)$, so $q|U_\alpha : U_\alpha \to V_\beta$ is surjective by Lemma 13.5. It is also open and injective (as $p|U_\alpha$ is injective). Thus $q|U_\alpha$ is a homeomorphism onto V_β, and $r|V_\beta = (p|U_\alpha) \circ (q|U_\alpha)^{-1}$ is a homeomorphism onto U. ∎

Theorem 13.7 (80.3)

If $p : E \to B$ is a covering map with E simply connected, then for any covering map $r : Y \to B$ there is a covering map $q : E \to Y$ with $r \circ q = p$.

Proof. Lemma 13.1 applies trivially and gives q with $r \circ q = p$; Lemma 13.6(a) shows q is a covering map. ∎

Lemma 13.8 (80.4)

If $p : E \to B$ is a covering map with E simply connected and $p(e_0) = b_0$, then b_0 has a neighbourhood U such that $i_* : \pi_1(U, b_0) \to \pi_1(B, b_0)$ is trivial.

Proof. Let U be evenly covered and $\tilde{U}$ the slice containing e_0. A loop f in U lifts to a loop in $\tilde{U}$, which is nulhomotopic in E; applying p gives a path homotopy in B from f to a constant. ■

Example 13.2

The infinite earring X (Example 11.2) is path connected and locally path connected but has no universal covering: for any neighbourhood U of the origin, some circle C_n lies in U, and C_n is a retract of X, so $\pi_1(C_n) \to \pi_1(X)$ is injective and $i_* : \pi_1(U) \to \pi_1(X)$ is nontrivial.

13.3 Covering transformations

An equivalence of p with itself is a **covering transformation**; they form the group $\mathcal{C}(E, p, B)$. The **normalizer** of H in G is $N(H) = \{g : gHg^{-1} = H\}$, the largest subgroup in which H is normal. Let $F = p^{-1}(b_0)$, $\Phi : \pi_1(B, b_0)/H_0 \to F$ the bijection induced by the lifting correspondence (Theorem 9.17), and $\Psi : \mathcal{C}(E, p, B) \to F$, $\Psi(h) = h(e_0)$, injective by uniqueness of liftings.

Lemma 13.9 (81.1)

The image of Ψ equals $\Phi(N(H_0)/H_0)$.

Proof. Let γ be the lifting of a loop α from e_0, $e_1 = \gamma(1) = \Phi([\alpha]H_0)$, $H_1 = p_*\pi_1(E, e_1)$. A covering transformation with $h(e_0) = e_1$ exists iff $H_1 = H_0$ (Theorem 13.2). Since $[\alpha]H_1[\alpha]^{-1} = H_0$ (Lemma 13.3), this holds iff $[\alpha]H_0[\alpha]^{-1} = H_0$, i.e. $[\alpha] \in N(H_0)$. ■

Theorem 13.10 (81.2)

$\Phi^{-1} \circ \Psi : \mathcal{C}(E, p, B) \to N(H_0)/H_0$ is a group isomorphism.

Proof. It is a bijection by Lemma 13.9. Let $h(e_0) = e_1$, $k(e_0) = e_2$, and choose paths γ, δ from e_0 to e_1, e_2; with $\alpha = p \circ \gamma$, $\beta = p \circ \delta$, $\Phi([\alpha]H_0) = e_1$ and $\Phi([\beta]H_0) = e_2$. Then $h \circ \delta$ runs from e_1 to $e_3 = h(k(e_0))$ and lifts β, so $\gamma * (h \circ \delta)$ lifts $\alpha * \beta$ from e_0 and ends at e_3: $\Phi([\alpha * \beta]H_0) = \Psi(h \circ k)$. ■

Corollary 13.11 (81.3)

H_0 is normal in $\pi_1(B, b_0)$ iff for all $e_1, e_2 \in p^{-1}(b_0)$ some covering transformation carries e_1 to e_2. Then $\mathcal{C}(E, p, B) \cong \pi_1(B, b_0)/H_0$.

Proof. If H_0 is normal, Ψ is surjective, giving h_i with $h_i(e_0) = e_i$; take $h_2 h_1^{-1}$. Conversely, transitivity makes Ψ surjective, so $N(H_0)/H_0 = \pi_1/H_0$ and $N(H_0) = \pi_1(B, b_0)$. ■

Corollary 13.12 (81.4)

If E is simply connected, $\mathcal{C}(E, p, B) \cong \pi_1(B, b_0)$.

If H_0 is normal, p is **regular**. Then $\mathcal{C}$ is transitive on every fibre $p^{-1}(b)$: for $x_1, x_2 \in p^{-1}(b)$ and a path β from b to b_0, lift β from x_1, x_2 to paths ending at e_1, e_2, and let $h(e_1) = e_2$; then h carries the first lifting to a lifting of β ending at e_2, which by uniqueness (lift $\bar{\beta}$ from e_2) is the second, so $h(x_1) = x_2$.

Example 13.3

Every covering of S^1 is regular; for $\mathbb{R} \to S^1$ the covering transformations are $x \mapsto x + n$. For the covering of the figure eight in Lemma 9.45, $\mathcal{C}$ is trivial: a loop around A lifts to a non-loop from the origin but to a loop from every other point of the fibre on the y-axis, and similarly for B, so every covering transformation fixes the origin.

For a group G of homeomorphisms of X, the **orbit space** X/G is the quotient by $x \sim g(x)$. The action is **properly discontinuous** if each x has a neighbourhood U with $g(U) \cap U = \emptyset$ for $g \neq e$; then $g_0(U) \cap g_1(U) = \emptyset$ for $g_0 \neq g_1$.

Theorem 13.13 (81.5)

Let X be path connected and locally path connected and G a group of homeomorphisms of X. The quotient map $\pi : X \to X/G$ is a covering map iff the action is properly discontinuous. In that case π is regular and G is its group of covering transformations.

Proof. π is open, since $\pi^{-1}\pi(U) = \bigcup_g g(U)$. If the action is properly discontinuous and U is as in the definition, $\pi^{-1}\pi(U)$ is the disjoint union of the open sets $g(U)$, each meeting each orbit at most once; $\pi|g(U)$ is a continuous open bijection onto $\pi(U)$, hence a homeomorphism. Conversely, if π is a covering, let U be the slice through x over an evenly covered neighbourhood of $\pi(x)$; if $g \neq e$ and $u' = g(u) \in U$ with $u \in U$, then $u' = u$ by injectivity of $\pi|U$; so g fixes u, and g is a covering transformation fixing a point, hence the identity (shown next), a contradiction. Each g is a covering transformation. If h is one with $h(x_1) = x_2$, then x_1, x_2 lie in one orbit, $g(x_1) = x_2$ for some g, and $h = g$ by Theorem 13.2. G is transitive on fibres, so π is regular (Corollary 13.11). ■

Theorem 13.14 (81.6)

If $p : X \to B$ is regular with group of covering transformations G, there is a homeomorphism $k : X/G \to B$ with $p = k \circ \pi$.

Proof. p is constant on orbits, so it induces a continuous k. Conversely p is a quotient map (continuous, open, surjective), and π is constant on each fibre of p since G is transitive on fibres; so π induces a continuous inverse $B \to X/G$. ■

Example 13.4

On the cylinder $X = S^1 \times I$, $h(x,t) = (-x,t)$ and $k(x,t) = (-x, 1-t)$ generate properly discontinuous actions of $\mathbb{Z}/2$; $X/\langle h \rangle$ is again a cylinder, $X/\langle k \rangle$ is the Möbius band. $\mathbb{Z}^2$ acting on $\mathbb{R}^2$ by translations gives the torus; $\{\pm 1\}$ acting on S^n gives P^n.

13.4 Existence of covering spaces

B is **semilocally simply connected** if each b has a neighbourhood U such that $\pi_1(U,b) \to \pi_1(B,b)$ is trivial. By Lemma 13.8 this is necessary for a universal covering.

Theorem 13.15 (82.1)

Let B be path connected, locally path connected and semilocally simply connected, $b_0 \in B$, and H a subgroup of $\pi_1(B, b_0)$. There is a covering map $p : E \to B$ and $e_0 \in p^{-1}(b_0)$ with $p_*\pi_1(E, e_0) = H$.

Proof. Step 1. Let $\mathcal{P}$ be the set of paths in B from b_0. Put $\alpha \sim \beta$ if $\alpha(1) = \beta(1)$ and $[\alpha * \bar{\beta}] \in H$, an equivalence relation; write $\alpha^{\#}$ for the class, E for the set of classes, and $p(\alpha^{\#}) = \alpha(1)$. Note: (1) $[\alpha] = [\beta]$ implies $\alpha^{\#} = \beta^{\#}$; (2) $\alpha^{\#} = \beta^{\#}$ implies $(\alpha * \delta)^{\#} = (\beta * \delta)^{\#}$ for any path δ from $\alpha(1)$, since $[(\alpha * \delta) * \overline{(\beta * \delta)}] = [\alpha * \bar{\beta}]$.

Step 2. For $\alpha \in \mathcal{P}$ and a path-connected neighbourhood U of $\alpha(1)$ let $B(U, \alpha) = \{(\alpha * \delta)^{\#} : \delta \text{ a path in } U \text{ from } \alpha(1)\}$; it contains $\alpha^{\#}$. If $\beta^{\#} \in B(U, \alpha)$, then $B(U, \beta) = B(U, \alpha)$: with $\beta^{\#} = (\alpha * \delta)^{\#}$, elements $(\beta * \gamma)^{\#} = (\alpha * (\delta * \gamma))^{\#}$ lie in $B(U, \alpha)$ by (2), and $\alpha^{\#} = (\beta * \bar{\delta})^{\#}$ gives the reverse inclusion. Hence if $\beta^{\#} \in B(U_1, \alpha_1) \cap B(U_2, \alpha_2)$, choosing a path-connected $V \subset U_1 \cap U_2$ containing $\beta(1)$, $B(V, \beta) \subset B(U_1, \beta) \cap B(U_2, \beta) = B(U_1, \alpha_1) \cap B(U_2, \alpha_2)$. So these sets form a basis.

Step 3. p is continuous ($B(U, \alpha) \subset p^{-1}(W)$ when $U \subset W$) and open ($p(B(U, \alpha)) = U$).

Step 4. Let U be a path-connected neighbourhood of b_1 with $\pi_1(U, b_1) \to \pi_1(B, b_1)$ trivial. Then $p^{-1}(U)$ is the union of the sets $B(U, \alpha)$ with $\alpha(1) = b_1$: if $p(\beta^{\#}) \in U$, choose δ in U from b_1 to $\beta(1)$ and $\alpha = \beta * \bar{\delta}$; then $\beta^{\#} = (\alpha * \delta)^{\#}$. Distinct such sets are disjoint by Step 2. And p is injective on $B(U, \alpha)$: if $(\alpha * \delta_1)(1) = (\alpha * \delta_2)(1)$, then $\delta_1 * \bar{\delta}_2$ is a loop in U at b_1, nulhomotopic in B, so $[\alpha * \delta_1] = [\alpha * \delta_2]$. Being a continuous open bijection, $p|B(U, \alpha)$ is a homeomorphism onto U.

Step 5. Let e_0 be the class of the constant path. For $\alpha \in \mathcal{P}$ and $c \in I$ let $\alpha_c(t) = \alpha(tc)$, and $\tilde{\alpha}(c) = \alpha_c^{\#}$. It lifts α from e_0 to $\alpha^{\#}$. It is continuous at c: a basic neighbourhood of $\tilde{\alpha}(c)$ is $B(U, \alpha_c)$, U a path-connected neighbourhood of $\alpha(c)$; for d near c, $\alpha([c, d]) \subset U$, and $[\alpha_d] = [\alpha_c * \delta]$ with δ the reparametrized restriction of α to $[c, d]$ (or $[d, c]$, reversed), so $\tilde{\alpha}(d) \in B(U, \alpha_c)$.

Step 6. E is path connected by Step 5, and locally path connected since it is locally homeomorphic to B; so p is a covering map.

Step 7. For a loop α at b_0, its lifting from e_0 ends at $\alpha^{\#}$, which equals e_0 iff $[\alpha] \in H$. By Theorem 9.17(c), $p_*\pi_1(E, e_0) = H$. ∎

Corollary 13.16 (82.2)

B has a universal covering space iff it is path connected, locally path connected and semilocally simply connected.

Proof. Necessity: Lemma 13.8 (each point of B is in the image of a point of E). Sufficiency: Theorem 13.15 with H trivial. ∎

Together with Theorem 13.4: for such B, equivalence classes of coverings correspond bijectively to conjugacy classes of subgroups of $\pi_1(B, b_0)$.